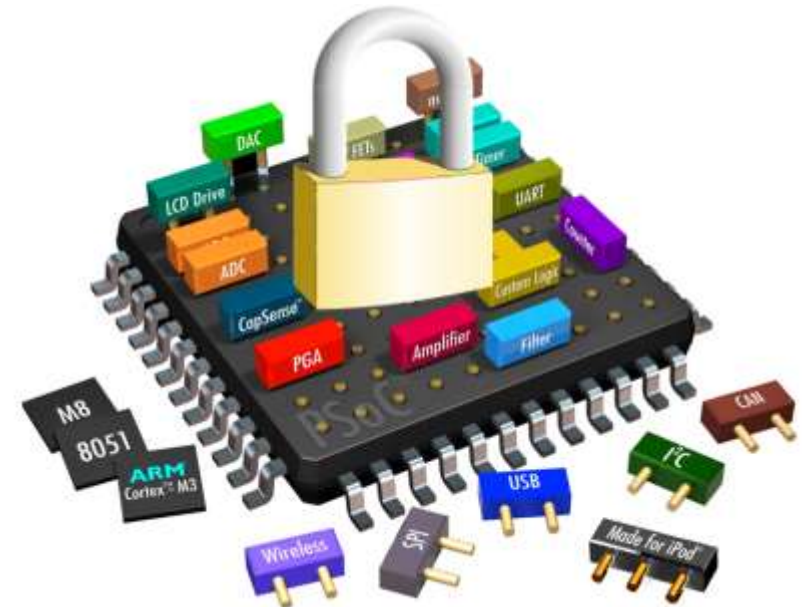


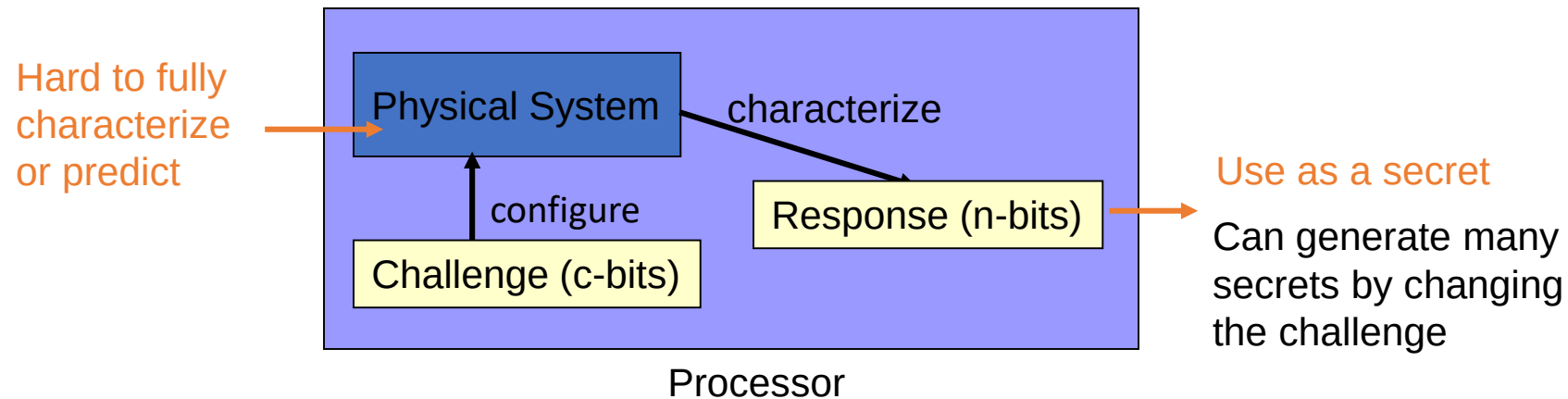
Hardware Security

Physical Unclonable Function (PUF)



Physical Unclonable Functions (PUFs)

- Generate keys from a **complex physical system**
- Security advantage
 - Keys are generated on demand → No non-volatile secrets
 - No need to program the secret
 - Can generate multiple master keys
- What can be hard to predict, but easy to measure?

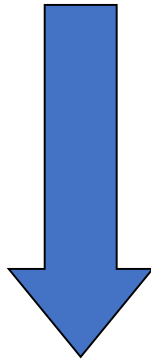


Definition

- A Physical Random Function or Physical Unclonable Function (PUF) is a function that is:
 - Based on a physical system
 - Easy to evaluate (using the physical system)
 - Its output looks like a random function
 - Unpredictable even for an attacker with physical access

WYSINWYG

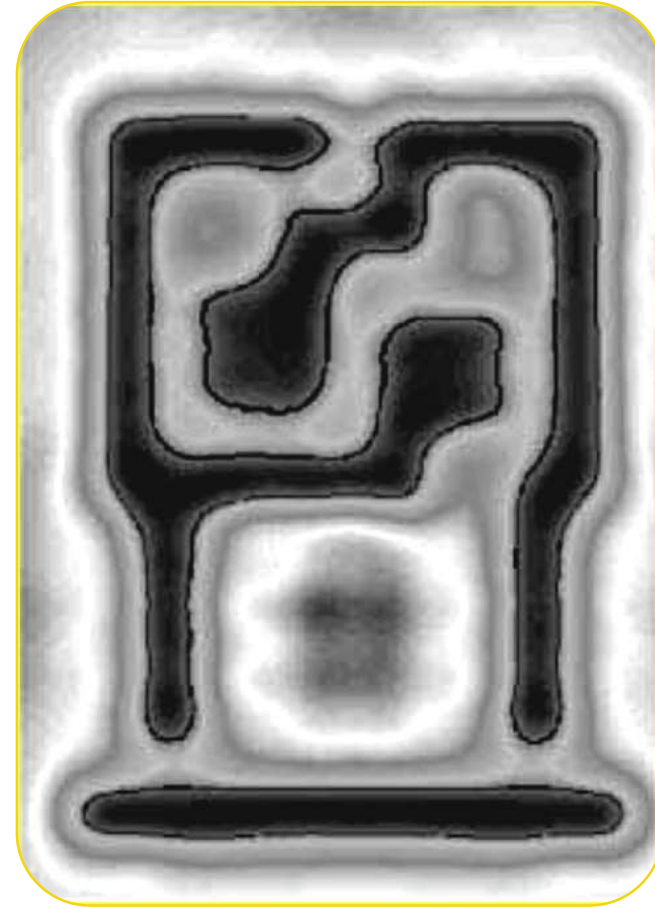
Sub-Wavelength **WYSINWYG**



What You See Is Not What You Get

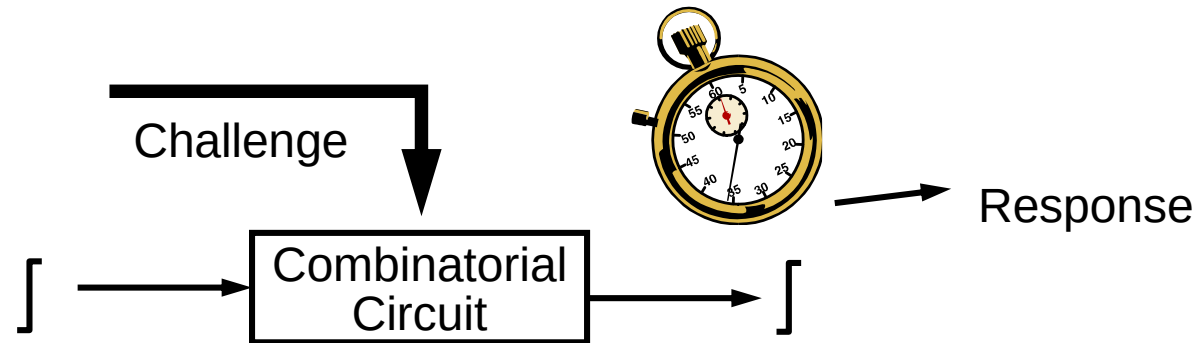
Process variations

No two transistors have the same parameters



Example: Delay PUF

- Because of process variations, no two integrated circuits are identical
- Experiments in which identical circuits with identical layouts were placed on different FPGAs show that path delays vary enough across ICs to use them for identification.



Physical Attacks

- Make PUF delays depend on overlaid metal layers and package
- Invasive attack (e.g., package removal) changes PUF delays and destroys PUF
- Non-invasive attacks are still possible
 - To find wire delays one needs to find precise relative timing of transient signals as opposed to looking for 0's and 1's
 - Wire delay is not a number but a function of challenge bits and adjacent wire voltages and capacitances